Tanush Sri Sai Petla

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.70
Sri Chaitanya JC, Vijayawada	Class XII (APBSIE)	2024	97.70%
Dr. KKR Gowtham Int'l Campus	Class X (CBSE)	2022	94.80%

RELEVANT COURSES

- **AI** : Foundations of Machine Learning, Deep Learning, AI for Humanity, Introduction to Modern AI and Classical AI, Programming for Artificial Intelligence.
- **CS** : Data Structures and Applications, Algorithms, DBMS, Discrete Mathematics, Introduction to Programming
- **Mathematics** : Probability and Random Variables, General Optimization, Convex Optimization, Metric Spaces, Numerical Analysis, Calculus
- **EE** : Control Systems, Linear Systems and Signal Processing, Matrix Theory

PROJECTS

Blind Bleedthrough Removal in Fluorescence Microscopy

(Ongoing Project under Prof. Ashesh, Dept. of AI, IITH)

- Developing a deep learning image-restoration framework to remove bleedthrough from real microscopy images.
- Modeling bleedthrough as a controllable degradation process and learning iterative restoration through synthetic mixing.
- Designing weak/self-supervised training strategies to recover structures without access to ground-truth targets.
- Exploring feature-decorrelation objectives to suppress channel interference and improve target reconstruction.

Mini Machine Learning Library & Autograd Engine (From Scratch)

- Developed a modular machine learning library from scratch in Python with a scikit-learn-style API.
- Implemented preprocessing, model selection, and classical machine learning algorithms from scratch.
- Built a reverse-mode autograd engine with neural network layers, loss functions, and SGD/Adam optimizers.
- Evaluated models on Spambase and Fashion-MNIST; custom MLP achieved 87.84% validation accuracy.

Deep Learning Framework & CNN Image Classifier

- Developed a modular CNN framework from scratch in NumPy with custom convolution, pooling, and activation layers.
- Implemented multi-channel convolution, pooling, flattening, and MLP modules for CIFAR-10 image classification.
- Trained CNN models in PyTorch using SGD, Momentum, RMSProp, BatchNorm, and Dropout techniques.
- Visualized activation maps and t-SNE embeddings while evaluating training and test performance.

Long-Context Sequence Modeling Framework

- Developed Vanilla RNN, LSTM, Transformer, and State Space Models for binary sentiment classification.
- Implemented self-attention, multi-head attention, positional encoding, and BPTT from scratch.
- Evaluated SGD and Adam optimizers with systematic ablation studies on sequence length and model design.
- Analyzed long-context learning using gradient flow, attention maps, and computational scaling behavior.

Neural Network Learning Framework

- Developed ADALINE and modular MLP models from scratch with configurable network architectures.
- Implemented SGD, Momentum, Adam, RMSProp, AdaGrad, Muon, and Nesterov optimizers.
- Evaluated depth, width, activations, and regularization through comprehensive ablation studies.
- Extracted neural features for kernel SVR and compared linear, polynomial, RBF, and neural kernels.

SKILLS

- **Programming Languages:** C++, C, Python, SQL
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, PyTorch
- **Tools & Platforms:** Git, GitHub, LaTeX, Jupyter Notebooks, Google Colab, VS Code
- **Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOPs), Database Management Systems, Machine Learning, Deep Learning

ACHIEVEMENTS

- Achieved a peak rating of **1529 (Specialist)** on Codeforces
- Achieved a peak rating of **1565** on Leetcode
- Solved **500+** DSA problems across **Codeforces, LeetCode, and CodeChef**.
- Achieved a **Global Rank of 870** in **Codeforces Round 1044 (Div. 2)**, competing against thousands of participants.

POSITION OF RESPONSIBILITIES

Events and Publicity Coordinator | SPICMAY IIT Hyderabad

- Coordinated the planning and execution of cultural events, workshops, and artist interactions with cross-functional teams.
- Managed publicity through posters, social media campaigns, and targeted student outreach across the campus.

Inspection Coordinator | MMC IIT Hyderabad

- Oversaw routine mess inspections to ensure hygiene, food quality, and compliance with established service standards.
- Coordinated with mess staff and student representatives to address concerns and improve dining services.